

Binomial Asset Pricing Model

Volume 1, Chapter 1

§1. Binomial Asset Pricing Model

§1.1. Notation

- **Time 0:** Very start of the period. **Time 1:** End of the first period.
- S_0 : Stock price at time 0. $S_1(H)$ is price at time 1 if coin flip is heads, $S_1(T)$ is price at time 1 if coin flip is tails.
- p is probability of heads, $q = 1 - p$ is probability of tails.
- r is the interest rate, where 1 dollar invested at time zero yields $1 + r$ at time one.
- K is the strike price for European call options.

§1.2. One-Period Binomial Model

Main Idea: Consider the following diagram.

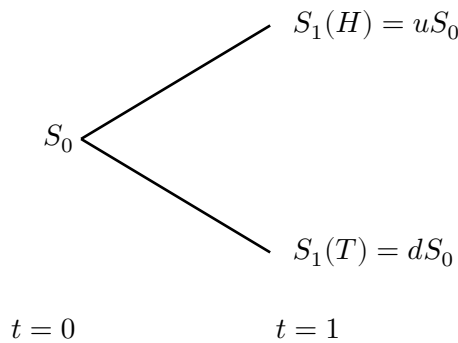


Figure 1.2.1: One-period binomial model

We know the outcome of the coin toss at time 1, but not time 0. Define

$$u = \frac{S_1(H)}{S_0}, \quad d = \frac{S_1(T)}{S_0}.$$

with $d < u$.

Definition 1.2.1 (Arbitrage).

A trading strategy that begins with no money, has zero probability of losing money, and positive probability of making money.

Remark 1.2.1.

We assume that there is no arbitrage in our analysis. Otherwise, we get contradictions because we can basically get infinite wealth generation.

Remark 1.2.2.

To prevent arbitrage, we must have $0 < d < 1 + r < u$.

Otherwise, if $d \geq 1 + r$, borrow money and buy the stock with the borrowed money, and the stock will always earn enough to pay off the borrowed money. If $u \leq 1 + r$, short the stock and use the proceeds in the money market, which will earn enough interest to buy back the stock at time 1. The converse is true.

Stock prices obviously do not move like this, but given a sufficient number of periods, this helps approximate continuous-time models.

Definition 1.2.2 (European Call Option).

A European call option gives its owner the right, but not the obligation, to buy one share of the stock at time one for strike price K at a certain point in time.

We consider the case where $S_1(T) < K < S_1(H)$. Observe that if we flip tails, the option expires worthless, but if we flip heads, the option has a payoff of $S_1(H) - K$.

Definition 1.2.3 (Arbitrage Pricing Theory).

Pricing options by replicating the option by trading in the stock and money markets.

Remark 1.2.3.

If we can get the same returns using the stock and money markets as we get from the option, then they should be valued at the same price.

Example 1.2.1.

Consider the following:

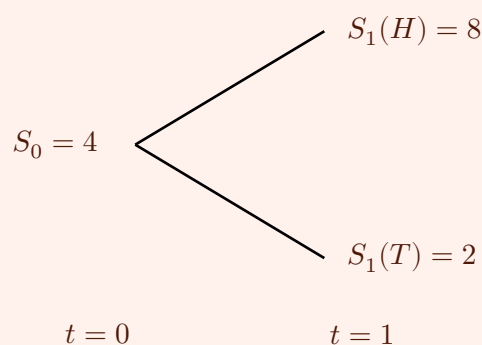


Figure 1.2.2: Example one-period binomial model

With $r = \frac{1}{4}$ and $K = 5$. Observe that if we have an initial wealth of $X_0 = 1.20$, borrow 0.80 to spend 2.00 on $\frac{1}{2}$ shares at time zero, then

$$X_1(H) = \frac{1}{2}S_1(H) - (1 + r)(0.80) = 3$$

and

$$X_1(T) = \frac{1}{2}S_1(T) - (1 + r)(0.80) = 0$$

which match the payoffs of the option! We have replicated the European call option, meaning that the option **must be worth the initial wealth** of 1.20.

If the option was worth more than 1.20, we could short the option and buy the replicating portfolio (i.e. buy $\frac{1}{2}$ shares and borrow 0.80). If the option was worth less than 1.20, we could buy the option and short the replicating portfolio (i.e. short sell $\frac{1}{2}$ shares and invest 0.80 in the money market). Both cases result in an **arbitrage opportunity**.

There are a few assumptions made:

- We can buy fractional shares.
- Interest rate for investing is the interest rate for borrowing.
- Purchase price of stock is the same as its selling price.
- Stock prices can be only one of two values in the next period.

Definition 1.2.4 (Derivative Security).

A derivative security is a security that pays, at time 1, $V_1(H)$ if the coin toss is heads and $V_1(T)$ if the coin toss is tails

To get the price of a derivative security at time 0, we replicate the security. We can derive a formula for this. Let X_0 be the value of the replicating portfolio, and hence, the **value of the option**. Let Δ_0 be the number of shares of stock we buy at time 0. The value of the replicating portfolio at time 1 is

$$X_1 = \underbrace{\Delta_0 S_1}_{\text{Position in stocks}} + \underbrace{(1+r)(X_0 - \Delta_0 S_0)}_{\text{Amount due on borrowed}} = \Delta_0(S_1 - (1+r)S_0) + (1+r)X_0$$

We want to find X_0 and Δ_0 so that $X_1(T) = V_1(T)$ and $X_1(H) = V_1(H)$.

Define $\tilde{p}$ and $\tilde{q} = 1 - \tilde{p}$ to be the values satisfying

$$S_0 = \frac{1}{1+r} [\tilde{p}S_1(H) + \tilde{q}S_1(T)]$$

This is similar to calculating the expected value for S_0 . Equating $X_1(H) = V_1(H)$ and $X_1(T) = V_1(T)$, we have

$$\begin{aligned} X_0 + \Delta_0 \left(\frac{1}{1+r} S_1(H) - S_0 \right) &= \frac{1}{1+r} V_1(H) \\ X_0 + \Delta_0 \left(\frac{1}{1+r} S_1(T) - S_0 \right) &= \frac{1}{1+r} V_1(T) \end{aligned} \quad (*)$$

We can add these equations together, multiplied by $\tilde{p}$ and $\tilde{q}$, to get

$$X_0 + \Delta_0 \left(\frac{1}{1+r} [\tilde{p}S_1(H) + \tilde{q}S_1(T)] - S_0 \right) = \frac{1}{1+r} [\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

Because of how we defined $\tilde{p}$ and $\tilde{q}$, we get

$$X_0 = X_0 + \Delta_0(S_0 - S_0) = \frac{1}{1+r} [\tilde{p}V_1(H) + \tilde{q}V_1(T)].$$

To solve for $\tilde{p}$, we have

$$S_0 = \frac{1}{1+r}(\tilde{p}uS_0 + (1-\tilde{p})dS_0)$$

$$1+r = \tilde{p}u + (1-\tilde{p})d$$

Which gives us

$$\tilde{p} = \frac{1+r-d}{u-d}.$$

Hence, we obtain

$$\tilde{q} = \frac{u-1-r}{u-d}.$$

To solve for Δ_0 , we subtract the equalities in * to obtain

$$\Delta_0 = \frac{v_1(H) - V_1(T)}{S_1(H) - S_1(T)}$$

Therefore, we should value the security at

$$V_0 = \frac{1}{1+r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)]$$

Definition 1.2.5 (Risk Neutral Probabilities).

We call $\tilde{p}$ and $\tilde{q}$ the risk-neutral probabilities.

Remark 1.2.4.

$\tilde{p}$ and $\tilde{q}$ are indeed probabilities as they sum to 1 and are non-negative. They represent the probabilities of the coin flip so that the **expected** growth rate of the stock is equal to r . In fact, because investors compensate for risk, if p and q represent the actual probabilities of the coin toss,

$$S_0 < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$

$$\frac{1}{1+r}[\tilde{p}S_1(H) + \tilde{q}S_1(T)] < \frac{1}{1+r}[pS_1(H) + qS_1(T)]$$

Remark 1.2.5.

We do not worry about the actual probabilities for valuation! We have constructed a replicating portfolio for the option, so the probabilities of up and down are irrelevant. Our replicating portfolio must earn the risk-free return.

§1.3. Multiperiod Binomial Model

What happens if we have multiple periods?

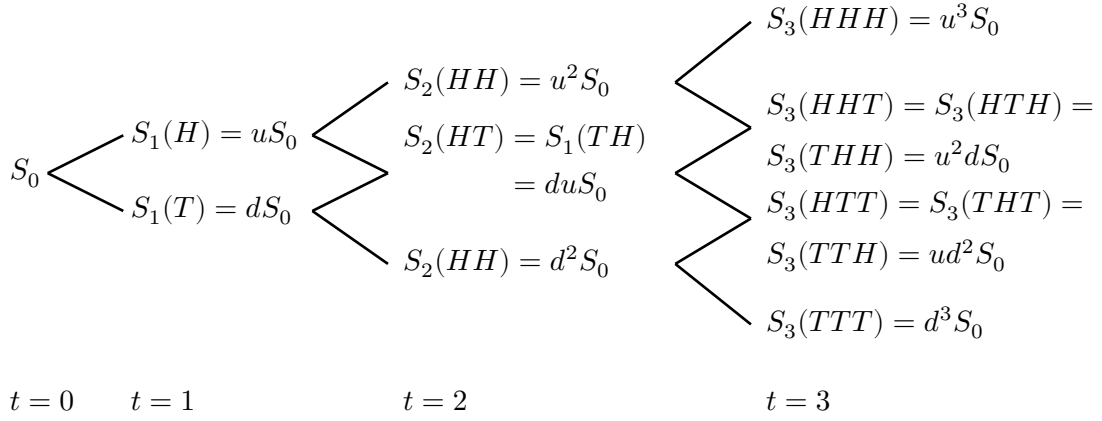


Figure 1.3.3: Three-period binomial model

To get the price using an N -period binomial model, we can work backwards. The idea is that at each time, we readjust the replicating portfolio/hedge to get risk-free returns. We can therefore extend the one-period case recursively.

More precisely, the wealth equation gives us the value of the replicating portfolio starting at time $t = 0$:

$$X_{n+1} = \underbrace{\Delta_n S_{n+1}}_{\text{Value of shares held since } t=n \text{ at new price}} + \underbrace{(1+r)(X_n - \Delta_n S_n)}_{\text{Money market investment}}.$$

To replicate a derivative with value V_n , we want to find the corresponding Δ_i (for all nodes of the tree) and value of X_i so that

$$X_n(\omega_1 \dots \omega_n) = V_n(\omega_1 \dots \omega_n)$$

for any outcomes ω_i . In other words, we look for a hedging strategy so that in the case of any outcome of up and down movements (the coin flips), **the value of the replicating portfolio is equal to the value of the derivative**. We get the following theorem by applying the one-period model:

Theorem 1.3.1 (Replication in the multiperiod binomial model).

Consider an N -period binomial asset-pricing model, with $0 < d < 1 + r < u$ (to prevent arbitrage), and define

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - 1 - r}{u - d}.$$

Let V_N be a random variable (representing the value of a derivative security's payoff at time N) which is dependent on the first N coin tosses $\omega_1 \omega_2 \dots \omega_n$. For $i = 0, 1, \dots, N - 1$, define

$$V_i(\omega_1 \dots \omega_i) = \frac{1}{1 + r} [\tilde{p} V_{i+1}(\omega_1 \dots \omega_i H) + \tilde{q} V_{i+1}(\omega_1 \dots \omega_i T)]$$

and also set

$$\Delta_i(\omega_1 \dots \omega_i) = \frac{V_{i+1}(\omega_1 \dots \omega_i H) - V_{i+1}(\omega_1 \dots \omega_i T)}{S_{i+1}(\omega_1 \dots \omega_i H) - S_{i+1}(\omega_1 \dots \omega_i T)}.$$

Forward recursively define

$$X_{i+1} = \Delta_i S_{i+1} + (1 + r)(V_i - \Delta_i S_i)$$

Then,

$$X_N(\omega_1 \dots \omega_N) = V_N(\omega_1 \dots \omega_N)$$

for all $\omega_1 \dots \omega_N$.

Definition 1.3.1 (Price of Derivative).

Given that the first n tosses are $\omega_1 \dots \omega_n$, define the price of the derivative security at time n to be $V_n(\omega_1 \dots \omega_n)$. Therefore, the price of the derivative security at time zero is V_0 .

Example 1.3.1.

Let $u = 2$, $d = \frac{1}{2}$, $N = 3$, $S_0 = 4$, and $r = \frac{1}{4}$. Then, $\tilde{p} = \tilde{q} = \frac{1}{2}$. Consider a lookback option that pays $V_3 = \max_{0 \leq n \leq 3} S_n - S_3$. Then,

$$\begin{aligned} V_3(HHH) &= S_3(HHH) - S_3(HHH) = 32 - 32 = 0 \\ V_3(HHT) &= S_3(HH) - S_3(HHT) = 16 - 8 = 8 \\ V_3(HTH) &= S_3(H) - S_3(HHT) = 8 - 8 = 0 \\ V_3(HTT) &= S_3(H) - S_3(HTH) = 8 - 2 = 6 \\ V_3(THH) &= S_3(THH) - S_3(HTT) = 8 - 8 = 0 \\ V_3(THT) &= S_3(TH) - S_3(THT) = 4 - 2 = 2 \\ V_3(TTH) &= S_0 - S_3(TTH) = 4 - 2 = 2 \\ V_3(TTT) &= S_0 - S_3(HHT) = 4 - 0.50 = 3.50 \end{aligned}$$

We get

$$\begin{aligned} V_2(HH) &= \frac{4}{5} \left[\frac{1}{2} V_3(HHH) + \frac{1}{2} V_3(HHT) \right] = 3.20 \\ V_2(HT) &= \frac{4}{5} \left[\frac{1}{2} V_3(HTH) + \frac{1}{2} V_3(HTT) \right] = 2.40 \\ V_2(TH) &= \frac{4}{5} \left[\frac{1}{2} V_3(THH) + \frac{1}{2} V_3(THT) \right] = 0.80 \\ V_2(TT) &= \frac{4}{5} \left[\frac{1}{2} V_3(TTH) + \frac{1}{2} V_3(TTT) \right] = 2.20 \end{aligned}$$

and

$$\begin{aligned} V_1(H) &= \frac{4}{5} \left[\frac{1}{2} V_2(HH) + \frac{1}{2} V_2(HT) \right] = 2.24 \\ V_1(T) &= \frac{4}{5} \left[\frac{1}{2} V_2(TH) + \frac{1}{2} V_2(TT) \right] = 1.20 \end{aligned}$$

giving us an option value of

$$V_0 = \frac{4}{5} \left[\frac{1}{2} V_1(H) + \frac{1}{2} V_1(T) \right] = 1.376.$$

Proof of Theorem: We prove by induction on n that $X_n(\omega_1 \dots \omega_n) = V_n(\omega_1 \dots \omega_n)$. The base case, $n = 0$, is true because we defined $X_0 = V_0$.

For the induction step, assume that our hypothesis holds true for some $n < N$. We show that it holds for $n + 1$. Fix a sequence of coin flips $\Omega = \omega_1 \dots \omega_{n+1}$. If $\omega_{n+1} = H$, then

$$X_{n+1}(\Omega H) = (1 + r)X_n(\Omega) - \Delta_n(\Omega)S_n(\Omega)(u - (1 + r)). \quad (*)$$

Our formula for Δ_n gives

$$\Delta_n = \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{S_{n+1}(\Omega H) - S_{n+1}(\Omega T)} = \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{uS_n(\Omega) - dS_n(\Omega)}$$

Substituting this into $(*)$, we get

$$\begin{aligned} X_{n+1}(\Omega H) &= (1 + r)X_n(\Omega) + \Delta_n S_n(\Omega)(u - (1 + r)) \\ &= (1 + r)V_n(\Omega) + \Delta_n S_n(\Omega)(u - (1 + r)) && \text{Induction Hypothesis} \\ &= (1 + r)V_n(\Omega) + \frac{V_{n+1}(\Omega H) - V_{n+1}(\Omega T)}{(u - d)S_n(\Omega)} S_n(\Omega)(u - (1 + r)) && \text{Substitution} \\ &= (1 + r)V_n(\Omega) + (V_{n+1}(\Omega H) - V_{n+1}(\Omega T))(\tilde{q}) && \tilde{q} = \frac{u - 1 - r}{u - d} \\ &= \tilde{p}V_{n+1}(\Omega H) + \tilde{q}V_{n+1}(\Omega T) + (V_{n+1}(\Omega H) - V_{n+1}(\Omega T))(\tilde{q}) && \text{Recursion for } V_n \\ &= \tilde{p}V_{n+1}(\Omega H) + \tilde{q}V_{n+1}(\Omega H) = V_{n+1}(\Omega H) \end{aligned}$$

as required. $\square$

§1.4. Computational Considerations

- If we compute naively for V_0 , where we do the full recursion for each node, the computational complexity is $\in O(2^N)$.
- Observe that at time n , there are only $n + 1$ possible stock prices. Therefore, it is reasonable to store intermediate results, reducing computational complexity.
- The idea is that for each time, starting from the back, we compute the valuation given a price, i.e. we find $v_{n(s)}$ for time n and stock price s , which should yield

$$v_{n(s)} = \frac{1}{1 + r} [\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)].$$

§2. Probability Theory on Coin Toss Space

§2.1. Finite Probability Spaces

We use **finite probability spaces** to model situations in a random experiment that permit **finitely many possible outcomes**.

Example 2.1.1.

If we toss a coin three times, the possible outcomes are

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

If the probability of a head is p (and thus the probability of a tail is $1 - p$), and we assume independent tosses, we get

$$\begin{aligned}\mathbb{P}(HHH) &= p^3, \mathbb{P}(HHT) = \mathbb{P}(HTH) = \mathbb{P}(THH) = qp^2, \\ \mathbb{P}(HTT) &= \mathbb{P}(THT) = \mathbb{P}(TTH) = q^2p, \mathbb{P}(TTT) = q^3\end{aligned}$$

Events are subsets of Ω .

Example 2.1.2.

The event “The first toss is a head” is the subset

$$\{\omega \in \Omega; \omega_1 = H\} = \{HHH, HHT, HTH, HTT\}$$

The probability of this event is

$$\begin{aligned}\mathbb{P}(\{\omega \in \Omega; \omega_1 = H\}) &= \mathbb{P}(HHH) + \mathbb{P}(HHT) + \mathbb{P}(HTH) + \mathbb{P}(THH) \\ &= p^3 + 2p^2q + pq^2 \\ &= p^2(p + q) + pq(p + q) \\ &= p^2 + pq \\ &= p\end{aligned}$$

More formally,

Definition 2.1.1 (Finite Probability Space, Event).

A finite probability space consists of a sample space Ω , which is a non-empty finite set, and a probability measure $\mathbb{P} : \Omega \rightarrow [0, 1]$ such that $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$.

An event is a subset of Ω . The probability of an event A is defined as $\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega)$

Remark 2.1.1.

Observe that

- $\mathbb{P}(\Omega) = 1$.
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$ if $A \cap B = \emptyset$.

§2.2. Random Variables

Definition 2.2.1 (Random Variables).

Let $(\Omega, \mathbb{P})$ be a finite probability space. A random variable is a function $\Omega \rightarrow \mathbb{R}$. We sometimes permit random variables to take the values $+\infty$ or $-\infty$.

We sometimes use ω to denote the argument of all the random variables, even if ω is a sequence.

Example 2.2.1.

Toss a coin three times. Define the random variables X = Total number of heads and Y = Total number of tails. Then,

$$\begin{aligned}X(HHH) &= 3 \\X(HHT) &= X(THH) = X(THH) = 2 \\X(HTT) &= X(THT) = X(TTH) = 1 \\X(TTT) &= 0\end{aligned}$$

and

$$\begin{aligned}Y(HHH) &= 0 \\Y(HHT) &= Y(THH) = Y(THH) = 1 \\Y(HTT) &= Y(THT) = Y(TTH) = 2 \\Y(TTT) &= 3.\end{aligned}$$

If each sequence of coin tosses has probability $\frac{1}{8}$ of occurring, then

$$\begin{aligned}\mathbb{P}(X = 0) &= P(TTT) = \frac{1}{8} \\ \mathbb{P}(X = 1) &= P(TTH) + P(THT) + P(HTT) = \frac{3}{8} \\ \mathbb{P}(X = 2) &= P(THH) + P(HTH) + P(HHT) = \frac{3}{8} \\ \mathbb{P}(X = 3) &= P(HHH) = \frac{1}{8}\end{aligned}$$

We write $X = j$ as shorthand for $\{\omega \in \Omega : X(\omega) = j\}$. We refer to the table of probabilities that X takes various values as the **distribution** of X . Distributions and random variables are different!

Definition 2.2.2 (Expected Values).

Let X be a random variable defined on finite probability space $(\Omega, \mathbb{P})$. The expectation/expected value of X is

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega)$$

When we use risk-neutral probabilities, we use the following notation:

$$\tilde{\mathbb{E}}X = \sum_{\omega \in \Omega} X(\omega) \tilde{\mathbb{P}}(\omega)$$

Definition 2.2.3 (Variance).

The variance of X is $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2]$

Remark 2.2.1.

Expectation is linear, i.e. $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$

Theorem 2.2.1 (Jensen's Inequality).

Let X be a random variable on a finite probability space and let $\varphi(x)$ be a convex function of a dummy variable x . Then

$$\mathbb{E}[\varphi(X)] \geq \varphi(\mathbb{E}X).$$

Proof: We first argue that

$$\varphi(x) = \max\{\ell(x); \ell \text{ is linear and } \ell(y) \leq \varphi(y) \text{ for all } y \in \mathbb{R}\}$$

i.e. $\varphi(x)$ is the maximum value attained at x of all linear functions that lie below it. It is clear that $\varphi(x) \geq \max(\dots)$ because we only consider linear functions that lie below φ . To see that $\varphi(x) \leq \max(\dots)$, fix x and consider the linear function ℓ that lies below φ but $\ell(x) = \varphi(x)$, which exists because φ is convex.

Let ℓ be a linear function below φ . Then,

$$\mathbb{E}[\varphi(X)] \geq \mathbb{E}[\ell(X)] = \ell(\mathbb{E}X)$$

by linearity of expectation. Because this is true for all ℓ , we get

$$\mathbb{E}[\varphi(X)] \geq \max\{\ell(\mathbb{E}X); \ell \text{ is linear and } \ell(y) \leq \varphi(y) \text{ for all } y \in \mathbb{R}\} = \varphi(\mathbb{E}X)$$

as required. $\square$

Example 2.2.2.

Because $\varphi(x) = x^2$ is convex, $\mathbb{E}[X^2] \geq (\mathbb{E}X)^2$.

§2.3. Conditional Expectations

Definition 2.3.1 (Conditional Expectation based on Information at Time n).

Let n satisfy $1 \leq n \leq N$ and let $\omega_1 \dots \omega_n$ be given and fixed. Let $\#H(\omega_{n+1} \dots \omega_N)$ be the number of heads in $\omega_{n+1} \dots \omega_N$ and let $\#T(\omega_{n+1} \dots \omega_N)$ be the number of tails in $\omega_{n+1} \dots \omega_N$. Define

$$\tilde{\mathbb{E}}_{n[X]}(\omega_1 \dots \omega_n) = \sum_{\omega_{n+1} \dots \omega_N} \tilde{p}^{\#H(\omega_{n+1} \dots \omega_N)} \tilde{q}^{\#T(\omega_{n+1} \dots \omega_N)} X(\omega_1 \dots \omega_N).$$

We call $\tilde{\mathbb{E}}_{n[X]}$ the conditional expectation of X based on the information at time n .

For the extreme cases, define $\tilde{\mathbb{E}}_0[X] = \tilde{\mathbb{E}}X$ and $\tilde{\mathbb{E}}_N[X] = X$.

Remark 2.3.1.

The conditional expectation is basically the value at a single node of our binomial tree. For example, if $\omega_1 = H$, then $\tilde{E}_1[X](\omega_1)$ is the value assuming that we took an up action in our tree at the very start.

Theorem 2.3.1 (Fundamental properties of conditional expectations).

Let $N \in \mathbb{Z}_+$. Let X and Y be random variables dependent on the first N coin tosses. Let $0 \leq n \leq N$ be given. The following properties hold.

1. **Linearity of conditional expectations.** For all constants c_1 and c_2 ,

$$\mathbb{E}_n[c_1X + c_2Y] = c_1\mathbb{E}_n[X] + c_2\mathbb{E}_n[Y].$$

2. **Taking out what is known.** If X only depends on the first n coin tosses,

$$\mathbb{E}_n[XY] = X\mathbb{E}_n[Y]$$

3. **Iterated conditioning.** If $0 \leq n \leq m \leq N$, then

$$\mathbb{E}_n[\mathbb{E}_m[X]] = \mathbb{E}_n[X].$$

This also implies that $\mathbb{E}[\mathbb{E}_m[X]] = \mathbb{E}X$

4. **Independence.** If X only depends on coin tosses $n + 1$ to N , then

$$\mathbb{E}_n[X] = \mathbb{E}X$$

5. **Conditional Jensen's inequality.** If $\varphi(x)$ is convex, then

$$\mathbb{E}_n[\varphi(X)] \geq \varphi(\mathbb{E}_n[X]).$$

§2.4. Martingales

We have so far used risk-neutral probabilities so that for every coin toss sequence, we have that $S_n(\omega_1 \dots \omega_n) = \frac{1}{1+r} [\tilde{p}S_{n+1}(\omega_1 \dots \omega_n H) + \tilde{q}S_{n+1}(\omega_1 \dots \omega_n T)] = \frac{1}{1+r} \tilde{\mathbb{E}}_n[S_{n+1}]$. We can divide by $(1+r)^n$ to get

$$\frac{S_n}{(1+r)^n} = \tilde{\mathbb{E}}_n \left[\frac{S_{n+1}}{(1+r)^{n+1}} \right] = \frac{1}{(1+r)^n} \tilde{\mathbb{E}}_n[S_{n+1}]$$

under the assumption that interest rates are non-random. This equation shows that the optimal forecast of the discounted stock price at time $n + 1$ is the discounted stock price at time n . This quirk motivates the definition of a martingale

Definition 2.4.1 (Martingale).

Consider the binomial asset-pricing model. Let $M_0, M_1, \dots, M_N$ be a sequence of random variables, with each M_n only depending on the first n coin tosses (observe that M_0 is a constant). Such a sequence of random variables is called an **adapted stochastic process**.

1. If

$$M_n = \mathbb{E}_n[M_{n+1}]$$

for $n = 0, \dots, N - 1$, this process is a **martingale**.

2. If

$$M_n \leq \mathbb{E}_n[M_{n+1}]$$

for $n = 0, \dots, N - 1$, this process is a **submartingale**.

3. 2. If

$$M_n \geq \mathbb{E}_n[M_{n+1}]$$

for $n = 0, \dots, N - 1$, this process is a **supermartingale**.

Remark 2.4.1.

If $M_0, \dots, M_N$ is a martingale and $n \leq N - 2$, then we have

$$M_{n+1} = \mathbb{E}_{n+1}[M_{n+2}]$$

Using the iterated conditioning probability, $\mathbb{E}_n[M_{n+1}] = \mathbb{E}_n[\mathbb{E}_{n+1}[M_{n+2}]] = \mathbb{E}_n[M_{n+2}]$. We can iterate this to show that

$$M_n = \mathbb{E}_n[M_m].$$

In fact, because $\mathbb{E}M_n = \mathbb{E}[\mathbb{E}_n[M_{n+1}]] = \mathbb{E}[M_{n+1}]$, we get

$$\mathbb{E}M_0 = \dots = \mathbb{E}M_N = M_0.$$

The main idea of a martingale is that given a sequence of events, the value of the state is equal to the expected value of the next state. Thus, the value of the martingale has **no tendency to rise or fall**. We find that stock prices with their actual probabilities for future prices are **submartingales**, i.e. their expected discounted prices at a future time are higher than their current prices, which makes sense given that investors usually enforce a risk premium.

Theorem 2.4.1 (Risk-Neutral Probabilities create Martingales).

Let

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - 1 - r}{u - d}$$

with $0 < d < 1 + r < u$. Under these risk-neutral probabilities, the discounted stock price is a martingale.

Proof: Make the observation that $\frac{S_{n+1}}{S_n}$ depends only on the $(n+1)$ st coin toss (it is either equal to u or d depending on the toss). With conditional probability properties, we get that

$$\begin{aligned}
\tilde{\mathbb{E}}_n \left[\frac{S_{n+1}}{(1+r)^{n+1}} \right] &= \tilde{\mathbb{E}}_n \left[\frac{S_n}{(1+r)^{n+1}} \cdot \frac{S_{n+1}}{S_n} \right] \\
&= \frac{S_n}{(1+r)^{n+1}} \tilde{\mathbb{E}}_n \left[\frac{S_{n+1}}{S_n} \right] && \text{Taking Out What is Known} \\
&= \frac{S_n}{(1+r)^{n+1}} \tilde{\mathbb{E}} \left[\frac{S_{n+1}}{S_n} \right] && \text{Independence} \\
&= \frac{S_n}{(1+r)^{n+1}} (\tilde{p}u + \tilde{q}d) \\
&= \frac{S_n}{(1+r)^n} \quad \square
\end{aligned}$$

Recalling our replicating portfolio idea, we see that $\Delta_0, \dots, \Delta_{N-1}$ and $X_0, \dots, X_{N-1}$ are both adapted stochastic process! In fact, there is something deeper about the wealth process:

Theorem 2.4.2 (Discounted Wealth Process is a Martingale).

Consider the binomial model with N periods. Let $\Delta_0, \dots, \Delta_{N-1}$ be an adapted portfolio process. Let $X_0 \in \mathbb{R}$ and let $X_1, \dots, X_N$ be generated by

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n).$$

Then $\frac{X_n}{(1+r)^n}$ for $n = 0, \dots, N$ is a martingale under the risk-neutral measure.

Proof: Observe

$$\begin{aligned}
\tilde{\mathbb{E}}_n \left[\frac{X_{n+1}}{(1+r)^{n+1}} \right] &= \tilde{\mathbb{E}}_n \left[\frac{\Delta_n S_{n+1}}{(1+r)^{n+1}} + \frac{X_n - \Delta_n S_n}{(1+r)^n} \right] \\
&= \tilde{\mathbb{E}}_n \left[\frac{\Delta_n S_{n+1}}{(1+r)^{n+1}} \right] + \tilde{\mathbb{E}}_n \left[\frac{X_n - \Delta_n S_n}{(1+r)^n} \right] && \text{Linearity} \\
&= \Delta_n \tilde{\mathbb{E}}_n \left[\frac{S_{n+1}}{(1+r)^{n+1}} \right] + \frac{X_n - \Delta_n S_n}{(1+r)^n} && \text{Taking out what is known} \\
&= \frac{\Delta_n S_n}{(1+r)^n} + \frac{X_n - \Delta_n S_n}{(1+r)^n} && \text{Theorem 2.4.1} \\
&= \frac{X_n}{(1+r)^n} \quad \square
\end{aligned}$$

Corollary 2.4.1.

$$\tilde{\mathbb{E}} \left[\frac{X_n}{(1+r)^n} \right] = X_0$$

Proof: Follows from Remark 2.4.1 as $\frac{X_n}{(1+r)^n}$ is a martingale.

Remark 2.4.2.

Our corollary shows that there is no arbitrage in the binomial model! Otherwise, suppose we

have a portfolio with $X_0 = 0$. If there was an arbitrage opportunity, $\tilde{\mathbb{E}}\left[\frac{X_n}{(1+r)^n}\right] > 0 = X_0$, a contradiction with our corollary.

Remark 2.4.3.

These ideas lead to the **First Fundamental Theorem of Asset Pricing**, which states that if there exists a risk-neutral measure in a model, i.e. a measure that agrees with the actual probabilities about the paths that have zero probability and where the risk-neutral discounted prices are martingales, then there is **no arbitrage**.

Theorem 2.4.3 (Risk-neutral pricing formula).

Consider N -period binomial asset-pricing model with $0 < d < 1 + r < u$ and risk-neutral probability measure $\tilde{\mathbb{P}}$. Let V_n be a random variable (the value of a derivative security with payoff at time N) dependent on the coin tosses. The price of the derivative security is then given by

$$V_n = \tilde{\mathbb{E}}_n \left[\frac{V_N}{(1+r)^{N-n}} \right]$$

and the discounted price of the derivative security is a martingale under $\tilde{\mathbb{P}}$, i.e.

$$\frac{V_n}{(1+r)^n} = \tilde{\mathbb{E}}_n \left[\frac{V_{n+1}}{(1+r)^{n+1}} \right], n = 0, \dots, N.$$

We define V_n here identically as in Theorem 1.3.1.

Proof: Exercise 8

Theorem 2.4.4 (Cash flow valuation).

Consider N -period binomial asset-pricing model with $0 < d < 1 + r < u$ and risk-neutral probability measure $\tilde{\mathbb{P}}$. Let $C_0, \dots, C_N$ be a sequence of random variables such that C_n depends only on $\omega_1 \dots \omega_n$. The price of a security at time n that pays $C_n, \dots, C_N$ at times $n, \dots, N$ respectively is

$$V_n = \tilde{\mathbb{E}}_n \left[\sum_{k=n}^N \frac{C_k}{(1+r)^{k-n}} \right], \quad n = 0, \dots, N.$$

The price process $V_n, n = 0, \dots, N$ satisfies

$$C_{n(\omega_1 \dots \omega_n)} = V_{n(\omega_1 \dots \omega_n)} - \frac{1}{1+r} [\tilde{p}V_{n+1}(\omega_1 \dots \omega_n H) + \tilde{q}V_{n+1}(\omega_1 \dots \omega_n T)]$$

Define

$$\Delta_n(\omega_1 \dots \omega_n) = \frac{V_{n+1}(\omega_1 \dots \omega_n H) - V_{n+1}(\omega_1 \dots \omega_n T)}{S_{n+1}(\omega_1 \dots \omega_n H) - S_{n+1}(\omega_1 \dots \omega_n T)}$$

for $n = 0, \dots, N-1$. If we set $X_0 = V_0$ and recursively define $X_1, \dots, X_N$ by

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - C_n - \Delta_n S_n)$$

we get

$$X_n(\omega_1 \dots \omega_n) = V_n(\omega_1 \dots \omega_n)$$

for all n and $\omega_1 \dots \omega_n$

Remark 2.4.4.

To finance practitioners, V_n is known as the **net present value** (NPV) of the sequence of payments $C_n, \dots, C_N$. Note that we include the payment at time n .

Proof: By induction on n . The base case is true by definition of X_0 . Suppose that $X_{n(\omega_1 \dots \omega_n)} = V_{n(\omega_1 \dots \omega_n)}$ for some $n = 0, \dots, N - 1$. We show that $X_{n+1}(\omega_1 \dots \omega_n H) = V_{n+1}(\omega_1 \dots \omega_n H)$ and a similar argument demonstrates that $X_{n+1}(\omega_1 \dots \omega_n T) = V_{n+1}(\omega_1 \dots \omega_n T)$, which in turn proves our theorem.

Because C_n depends on the first n coin tosses only, we can modify the equation for V_n :

$$\begin{aligned} V_n &= \tilde{\mathbb{E}}_n \left[\sum_{k=n}^N \frac{C_k}{(1+r)^{k-n}} \right] \\ &= C_n + \tilde{\mathbb{E}}_n \left[\frac{1}{1+r} \sum_{k=n+1}^N \frac{C_k}{(1+r)^{k-(n+1)}} \right] \\ &= C_n + \tilde{\mathbb{E}}_n \left[\frac{1}{1+r} \tilde{\mathbb{E}}_{n+1} \left[\sum_{k=n+1}^N \frac{C_k}{(1+r)^{k-(n+1)}} \right] \right] \\ &= C_n + \tilde{\mathbb{E}}_n \left[\frac{1}{1+r} V_{n+1} \right] \end{aligned}$$

This leads to the following:

$$\begin{aligned} V_n - C_n &= \tilde{\mathbb{E}}_n \left[\frac{1}{1+r} V_{n+1} \right] \\ &= \frac{\tilde{p}V_{n+1}(\omega_1 \dots \omega_n H) + \tilde{q}V_{n+1}(\omega_1 \dots \omega_n T)}{1+r} \end{aligned}$$

For the rest of the proof, we suppress $\omega_1 \dots \omega_n$ for ease of writing. Observe that

$$\begin{aligned} X_{n+1}(H) &= \Delta_n S_{n+1}(H) + (1+r)(X_n - C_n - \Delta_n S_n) \\ &= \frac{V_{n+1}(H) - V_{n+1}(T)}{S_{n+1}(H) - S_{n+1}(T)} (S_{n+1}(H) - (1+r)S_n) + (1+r)(V_n - C_n) \\ &= \frac{V_{n+1}(H) - V_{n+1}(T)}{S_n(u-d)} (S_n)(u-1-r) + \tilde{p}V_{n+1}(H) + \tilde{q}V_{n+1}(T) \\ &= (V_{n+1}(H) - V_{n+1}(T))\tilde{q} + \tilde{p}V_{n+1}(H) + \tilde{q}V_{n+1}(T) \\ &= V_{n+1}(H) \end{aligned}$$

as required. $\square$

§2.5. Markov Processes

Markov processes are a way to simplify calculations by determining what is relevant and what is not, akin to the computational “shortcuts” discussed in 1.4.

Definition 2.5.1 (Markov Process).

Consider the binomial asset-pricing model. Let $X_0, \dots, X_N$ be an adapted process. If, for every $n \in \{0, \dots, N-1\}$ and function $f(x)$, there is another function $g(x)$ (which we get given n and f), such that

$$\mathbb{E}_n[f(X_{n+1})] = g(X_n)$$

we say that $X_0, \dots, X_N$ is a Markov process.

In a bit, we prove that if such a g exists, then the payoff of a derivative security is random only through its dependence on X_N , then we can create a pricing algorithm that does not need to store path information.

Example 2.5.1.

In the binomial model, where

$$S_{n+1}(\omega_1 \dots \omega_{n+1}) = \begin{cases} uS_n(\omega_1 \dots \omega_n) & \text{if } \omega_{n+1} = H \\ dS_n(\omega_1 \dots \omega_n) & \text{if } \omega_{n+1} = T \end{cases}$$

we get that

$$\mathbb{E}_n[f(S_{n+1})](\omega_1 \dots \omega_n) = pf(uS_n(\omega_1 \dots \omega_n)) + qf(dS_n(\omega_1 \dots \omega_n)).$$

The function $g(x) = pf(ux) + qf(dx)$ shows that this process is Markov.

To see why Markov processes are nice, consider the valuation of a derivative security with a payoff at time N is $v_N(S_N)$ for some function v_N . Then, the price at time n is given by

$$V_n = \frac{1}{1+r} \tilde{\mathbb{E}}_n[V_{n+1}], n = 0, \dots, N-1.$$

But $V_N = v_N(S_N)$ and the stock price process is Markov, so

$$V_{N-1} = \frac{1}{1+r} \tilde{\mathbb{E}}_{N-1}[v_N(S_N)] = v_{N-1}(S_{N-1})$$

for some function v_{N-1} . We can continue generating such v_n to get $V_n = v_n(S_n)$. Using the definition of $\tilde{\mathbb{E}}$, we get

$$v_n(s) = \frac{1}{1+r} [\tilde{p}v_{n+1}(us) + \tilde{q}v_{n+1}(ds)]$$

for $n = 0, \dots, N-1$!

Remark 2.5.1.

Martingales are not necessarily Markov processes and Markov processes are not necessarily Martingales. For example, non-discounted stock prices are Markov processes but they are not Martingales, unless $pu + qd = 1$.

Lemma 2.5.1 (Independence).

Consider the N -period binomial asset pricing model with $n \in \{0, \dots, N\}$. Suppose the random variables $X^1, \dots, X^K$ depend only on coin tosses 1 through n and the random variables $Y^1, \dots, Y^L$ depend only on coin tosses $n+1, \dots, N$. Let $f(x^1, \dots, x^K, y^1, \dots, y^L)$ be a function of dummy variables $x^1, \dots, x^K$ and $y^1, \dots, y^L$, and define

$$g(x^1, \dots, x^K) = \mathbb{E}f(x^1, \dots, x^K, Y^1, \dots, Y^L).$$

Then,

$$\mathbb{E}_n[f(X^1, \dots, X^K, Y^1, \dots, Y^L)] = g(X^1, \dots, X^K).$$

If $K = L = 1$, this means if we set $g(x) = \mathbb{E}f(x, Y)$, then $\mathbb{E}_n[f(X, Y)] = g(X)$. This is the analog of “taking out what is known”.

Proof: Let $\omega_1 \dots \omega_n$ be fixed but arbitrary. By the definition of conditional expectation,

$$\mathbb{E}_n[f(X, Y)](\omega_1 \dots \omega_n) = \sum_{\omega_{n+1} \dots \omega_N} f(X(\omega_1 \dots \omega_n), Y(\omega_{n+1} \dots \omega_N)) p^{\#H(\omega_{n+1} \dots \omega_N)} q^{\#T(\omega_{n+1} \dots \omega_N)}.$$

On the other hand,

$$g(x) = \mathbb{E}_n[f(x, Y)](\omega_1 \dots \omega_n) = \sum_{\omega_{n+1} \dots \omega_N} f(x, Y(\omega_{n+1} \dots \omega_N)) p^{\#H(\omega_{n+1} \dots \omega_N)} q^{\#T(\omega_{n+1} \dots \omega_N)}.$$

Which shows that

$$\mathbb{E}_n[f(X, Y)](\omega_1 \dots \omega_n) = g(X(\omega_1 \dots \omega_n)).$$

Example 2.5.2.

The maximum-to-date derivative is not a Markov process! $M_2(TH) = M_2(TT)$ but $M_3(TTH) = 4 \neq M_3(THH) = 8$ (which leads us to the discovery that $\mathbb{E}_3[M_3](TH)$ cannot equal $\mathbb{E}_3[M_3](TT)$).

Markov processes are much easier to compute. Is it possible to turn non-Markov processes Markov? The general idea is to add “state variables” to recover the Markov property.

Definition 2.5.2 (Multi-dimensional Markov).

Consider binomial asset-pricing model. Let $\{(X_n^1, \dots, X_n^K); n = 0, \dots, N\}$ be a K -dimensional adapted process; i.e., K one-dimensional adapted processes. If, for every n between 0 and $N - 1$ and for every function $f(x^1, \dots, x^K)$, there is another function $g(x^1, \dots, x^K)$ such that

$$\mathbb{E}_n[f(X_n^1, \dots, X_n^K)] = g(X_n^1, \dots, X_n^K),$$

we say that $\{(X_n^1, \dots, X_n^K); n = 0, \dots, N\}$ is a K -dimensional Markov process.

Example 2.5.3.

We can now turn the maximum-to-date derivative process Markov. Consider the two-dimensional

adapted process $\{(S_n, M_n); n = 0, \dots, N\}$ where S_n is the stock price and M_n is the stock price maximum-to-date.

Let $Y = \frac{S_{n+1}}{S_n}$, only dependent on the $(n + 1)$ st coin flip. We want to compute

$$\mathbb{E}_n[f(S_{n+1}, M_{n+1})] = \mathbb{E}_n[f(S_n Y, \max(M_n, (S_n, Y)))].$$

If we let

$$g(s, m) = pf(us, \max(m, us)) + qf(ds, \max(m, ds))$$

We have found a g so that $g(s, m) = \mathbb{E}[f(S_{n+1}, M_{n+1})]$.

Remark 2.5.2.

The Markov property is a “one-step-ahead” property. I.e. the Markov property implies that if $n \leq N - 2$, then for every function h there exists a function f so that

$$\mathbb{E}_{n+1}[h(X_{n+2})] = f(X_{n+1}).$$

This leads us to

$$\mathbb{E}_n[h(X_{n+2})] = \mathbb{E}_n[\mathbb{E}_{n+1}[h(X_{n+2})]] = \mathbb{E}_n[f(X_{n+1})]$$

or

$$\mathbb{E}_n[h(X_{n+2})] = g(X_n).$$

The upshot is that if $0 \leq n \leq m \leq N$ and $h(x^1, \dots, x^K)$ is a function, there exists another function $g(x^1, \dots, x^K)$ so that

$$\mathbb{E}_n[h(X_m^1, \dots, X_m^K)] = g(X_m^1, \dots, X_m^K)$$

Using the “one-step-ahead” property, we can simplify calculations for the valuation of derivatives. The idea is that we are guaranteed to have a function v_n such that

$$\tilde{\mathbb{E}}_n \left[\frac{V_N}{(1+r)^{N-n}} \right] (\omega_1 \dots \omega_n) = v_n(X_n(\omega_1 \dots \omega_n)) \text{ for all } \omega_1 \dots \omega_n,$$

meaning that

$$V_n = v_n(X_n)$$

i.e. there exists a function to calculate the value of the derivative. This is summarized in the following theorem:

Theorem 2.5.1 (Markov Valuation).

Let $X_0, \dots, X_N$ be a Markov process under risk-neutral probability measure $\tilde{\mathbb{P}}$ in the binomial model. Let $v_N(x)$ be a function and consider a security whose payoff at time N is $v_N(X_N)$. Then, for each $n = 0, \dots, N$, there exists a function v_n of X_n such that

$$V_n = v_n(X_n), n = 0, \dots, N.$$

There is a recursive algorithm to compute such v_n .

Example 2.5.4.

Going back to the max-lookback stock option, we have

$$V_n = \frac{1}{1+r} \tilde{\mathbb{E}}_n [v_{n+1}(S_{n+1}, M_{n+1})] = \frac{1}{1+r} \tilde{\mathbb{E}}_n \left[v_{n+1} \left(S_n \cdot \frac{S_{n+1}}{S_n}, \max \left(M_n, S_n \cdot \frac{S_{n+1}}{S_n} \right) \right) \right].$$

Then,

$$v_n(s, m) = \frac{1}{1+r} \tilde{\mathbb{E}}_n \left[v_{n+1} \left(S_n \cdot \frac{S_{n+1}}{S_n}, \max \left(M_n, S_n \cdot \frac{S_{n+1}}{S_n} \right) \right) \right]$$

by the Markov property of the max-lookback stock option. We get

$$v_n(s, m) = \frac{1}{1+r} [\tilde{p}v_{n+1}(us, \max(m, us)) + \tilde{q}v_{n+1}(ds, m)]$$

Because $ds < m$ if $d \leq 1$ (usually assumed), giving us a recursive formula for v_n .

§3. State Prices

§3.1. Change of Measure

So far, we have encountered the actual probabilities and the risk-neutral probabilities. Actual probabilities are useful because they reflect the real world. Risk-neutral probabilities are useful because under this measure, the discounted price of assets are martingales, helping us solve systems of equations.

Definition 3.1.1 (Radon-Nikodym Derivative).

Let Ω be a finite sample space with probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$. Assume that in both $\mathbb{P}$ and $\tilde{\mathbb{P}}$, every element of Ω has positive probability. Define

$$Z(\omega) = \frac{\tilde{\mathbb{P}}(\omega)}{\mathbb{P}(\omega)}$$

Observe that Z is a random variable! We call Z the Radon-Nikodym derivative of $\tilde{\mathbb{P}}$ with respect to $\mathbb{P}$.

Theorem 3.1.1 (Properties of Z).

1. $\mathbb{P}(Z > 0) = 1$
2. $\mathbb{E}Z = 1$
3. $\tilde{\mathbb{E}}Y = \mathbb{E}[ZY]$ for any random variable Y